Diego Solorzano

(951)463-3329 | dasolorz@uci.edu | linkedin.com/in/diegoasolo/

Education

University of California, Irvine **Expected June 2026**
B.S., Electrical Engineering, Specialization in Semiconductors and Optoelectronics GPA: 3.755/4.0
Campus Wide Honors Collegium
Honors Thesis: Enhancing Defect Emission in hBN via Gold Nanorods (In Progress) Advisor: Maxim Shcherbakov

Research Experience

Shcherbakov Nanophotonics Lab February 2023 – Present
Undergraduate Research Assistant *Irvine, CA*

- Conducting an independent honors research project enhancing quantum emission from defect centers in hexagonal Boron Nitride (hBN) using **plasmonic cavities**
- Designed and assembled a custom **photoluminescence (PL)** measurement setup, including laser excitation, optical alignment, and fiber-coupled spectrometer integration to characterize emission properties of photonic devices
- Performed **FDTD simulations** using **Ansys Lumerical** to model the optical response of 2D semiconductor heterostructure devices, leveraging simulation data to enhance device PL emission
- Devised an automated PMMA-assisted wet transfer setup that increased process efficiency and improved the optical performance of 2D heterostructure devices
- Developed a procedure for drop casting nano-particles onto devices and characterized their distribution using **Scanning electron Microscopy (SEM)**

HyperXite at UCI July 2024 – Present
Mag-Lev R&D Engineer *Irvine, CA*

- Pioneered the formation of a magnetic levitation sub-team, initiating development of the first Electrodynamic Suspension (EDS) prototypes utilizing a linear Halbach array of permanent magnets
- Conducted **COMSOL Multiphysics** simulations of EDS systems to evaluate and optimize permanent magnet Configurations, analyzing lift and drag forces to achieve a 50% increase in the lift-to-drag ratio
- Designed and built an experimental test rig and scaled prototype to validate simulations with on board distance sensors, achieving a maximum levitation height of 25 mm

Professional Experience

Menlo Microsystems, Inc June 2024 – September 2024
Quality and Failure Analysis Engineering Intern *Irvine, CA*

- Implemented JEDEC standard tests, including Moisture Sensitivity Level, Temperature Cycling, and High Temperature Storage life, verifying component integrity under various conditions
- Completed verification testing using National Instruments Semiconductor Test System and cycling tests at room and high temperatures to ensure reliability and validate stress test performance
- Conducted comprehensive **failure analysis**, of **MEMS devices** including manual probing for resistance and leakage, surface analysis with optical microscopy and SEM, and destructive analysis such as cross-sectional examination, mechanical/chemical de-capsulation, and red dye analysis

HyperXite at UCI Jan 2024 - June 2024
Hardware Engineering Intern *Irvine, CA*

- Utilized **Altium** to design a 4-layer control board PCB that interfaces multiple temperature, pressure, and motion sensors, along with the pneumatic braking system, to the onboard Raspberry Pi
- Leveraged **LTspice** to simulate and analyze idealized circuit designs, verifying signal integrity, power distribution, and operational stability to meet control board specifications
- Integrated a low-voltage and high-voltage system consisting of a battery management system (BMS), control board, power converters, and batteries onto the pod

Teaching Experience

UCI School of Physical Science — Sep 2023 – Jun 2024
Learning Assistant — *Irvine, CA*

- Providing dedicated assistance in class to students across three physics courses—Mechanics, Electromagnetism, and Waves & Optics—by reinforcing key concepts and facilitating classroom discussions
- Partnering closely with the Teaching Assistant and Professor to organize and assist engaging class sessions by going over learning goals and identifying common weak points among students
- Effectively communicated complex physics concepts in a clear and accessible manner, tailoring explanations to meet the diverse learning styles of students
- Earned a Learning Assistant certification demonstrating proficiency in implementing multiple evidence-based pedagogical methods

Projects

Fog Hacks | *Project Manager* | *Senior Design* — July 2025 – Present

- Leading a multidisciplinary team to develop a novel system that projects clear images through fog using computational optics and a **digital micro-mirror device (DMD)** for dynamic phase correction
- Simulating an optical pipeline with **python**, integrating phase mask optimization, ray projection and scattering modeling, and image quality assessment to enhance visibility through turbid media
- Constructing the optical set up featuring a laser, beam expanding optics, DMD, fog chamber, and image sensor

Quantum Error Correction Codes — April 2025 – June 2025

- Developed and implemented the 9-qubit Shor error-correcting code using **Qiskit** on IBM Quantum hardware, designing and optimizing quantum circuits to encode, decode, and correct arbitrary single-qubit errors
- Extended 9-qubit Shor error correction to a multi-error implementation using 13 qubits in Qiskit, enabling sequential detection and correction of bit-flip (X) and phase-flip (Z) errors within a single circuit

Autonomous Recruitment Rover | *Project Manager* | *Regional Competition Winner* — Sep 2024 – April 2025

- Directed a 17-member interdisciplinary team in designing and building an autonomous rover, presenting and demoing our results at a regional club conference
- Led weekly general and sub-team meetings, coordinating progress updates, aligning technical milestones with overall project goals, and managing budget allocation and spending
- Designed a custom 16V battery in a 4s8p configuration with a dedicated BMS to provide sufficient power to high and low voltage systems, leading to a 12% increase in power efficiency

Presentations

"Plasmonic Nanocavity-boosted Quantum Emission in Few-layer Hexagonal Boron Nitride," Poster, Undergraduate Research Opportunities Program Symposium, May 2025

"HyperXite: Study and Design of Electrodynamic suspension," Oral Presentation, Hyperloop Global Competition, May 2025

"Autonomous Recruitment Rover: Overivew and Demo," Oral Presentation, Theta Tau Southwest Regional Conference, April 2025

Fellowships and Awards

Hyperloop Global: 1st Place Levitation System Design (Associated with HyperXite at UCI)	May 2025
UROP Travel Award (Awarded $1,000, Associated with HyperXite at UCI)	May 2025
Theta Tau: Southwest Region Engineering Competition Winner	April 2025
Eddleman Quantum Institute: Quantum UROP Fellowship (Awarded $3,000)	Jun 2024

Extracurricular and Leadership

Theta Tau: Professional Co-Ed Engineering Fraternity April 2023 – Present

Director of Engineering *August 2024 - June 2025*

- Led the planning and technical oversight of project initiatives within the organization, coordinating interdisciplinary teams to design and prototype engineering projects for recruitment and community engagement

Director of Media and Chapter History *August 2023 – June 2024*

- Developed graphic design skills to create digital and print media for public events, while documenting and preserving chapter activities

Skills

Lab Skills: Optical set up design & alignment, PL spectroscopy, 2D material wet transfer, Scanning Electron Microscopy, Atomic Force Microscopy, Digital Multimeter, Oscilloscope, Signal generator, Soldering

Software: ANSYS Lumerical, COMSOL, Altium, SolidWorks, Blender, Adobe Illustrator

Languages: Python, Matlab, C/C++, LaTeX

Hardware: Raspberry Pi, Arduino, ESP32